

RLHF objective

$$\max \mathbb{E}[r] - \beta \text{KL}$$

start

closed-form optimum

$$\pi_r \propto \pi_{\text{ref}} e^{r/\beta}$$

step 1

invert for reward

$$r = \beta \log \frac{\pi_r}{\pi_{\text{ref}}} + \beta \log Z$$

step 2

$Z(x)$ cancels

differences kill $\log Z$

step 3

DPO loss

$$-\log \sigma(\beta \Delta \log\text{-ratio})$$

step 4